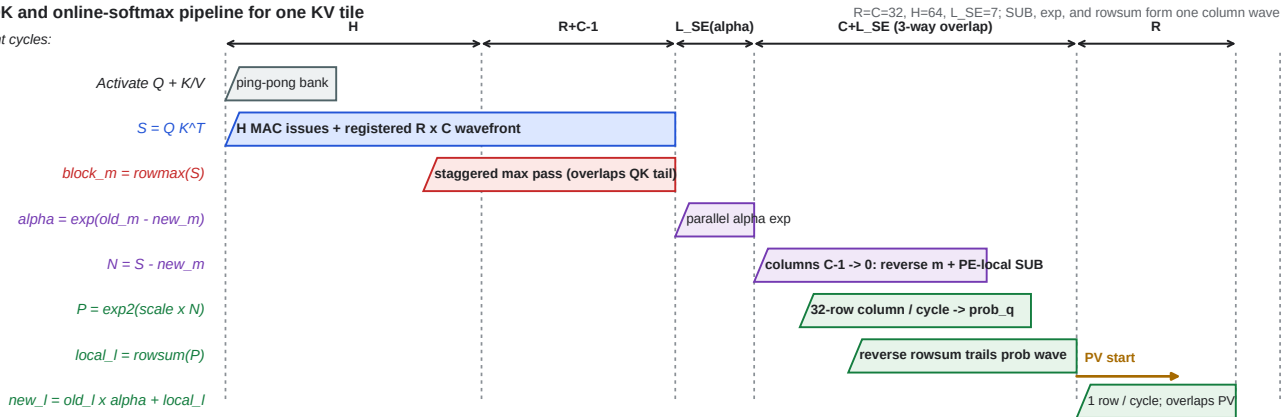


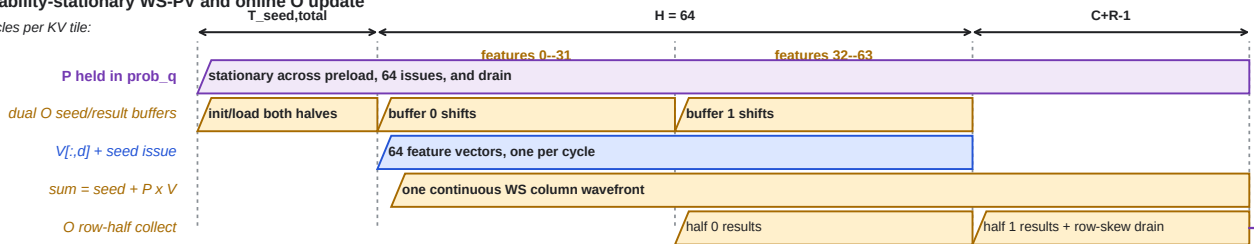
a QK and online-softmax pipeline for one KV tile

Dominant cycles:



b Probability-stationary WS-PV and online O update

Dominant cycles per KV tile:



T_seed,total=1 for j=0; approximately 4R for j>0. j<last repeats (a)-(b) with retained (m,l,O); j=last normalizes and writes back.

last KV: normalize + WB